

VIVA PREPARATION GUIDE

Enterprise Network Security Design Using Cisco ASA Firewall

NAT & PAT Configuration, Access Control Lists, and
DNS Integration in a Simulated Enterprise Network

Prepared for: Project Viva / Defense
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1. Basic Network Understanding

Q: What is the "inside" network in this project?

- The trusted, private company network
- Address range: 192.168.1.0/24
- Contains: PCs, Internal Web Server
- Security level on ASA: 100 (highest = most trusted)

Q: What is the "outside" network?

- The untrusted network — represents the internet/ISP side
- Address range: 6.6.6.0/24 (between ASA and ISP router)
- Security level on ASA: 0 (lowest = least trusted)

Q: What is the 5.5.5.0/24 network?

- A separate network beyond the ISP router
- Represents "the internet" where remote/public servers live
- Contains: DNS Server (5.5.5.2), Outside Web Server (5.5.5.3)

Q: Why does the ASA have a security-level for each interface?

- ASA uses security levels to decide default traffic behavior
- Higher → Lower security level = allowed by default (inside → outside)
- Lower → Higher security level = blocked by default (outside → inside), unless an ACL explicitly permits it
- This is the core rule that makes firewalls "secure by default"

Q: Why is inside security-level 100 and outside security-level 0?

- 100 = fully trusted (internal company devices)
- 0 = fully untrusted (public internet side)
- This gap enforces that outside can't reach inside unless specifically allowed

2. IP Addressing and Devices

Q: What IP does the ASA inside interface have?

- 192.168.1.1 (this is also the default gateway for all inside devices)

Q: What IP does the ASA outside interface have?

- 6.6.6.2

Q: Why does PC0 have a static IP but PC1/PC2 use DHCP?

- To demonstrate both addressing methods in one project
- Static IP: manually fixed, doesn't change (used for PC0 in this project, but normally used for servers)
- DHCP: automatically assigned by the ASA's DHCP server, used for regular end-user PCs

Q: Why should servers (Web Server, DNS Server) always use static IPs, not DHCP?

- Servers must be reachable at a fixed, predictable address
- NAT rules, ACLs, and DNS records all point to a fixed IP
- If a server's IP changed (as could happen with DHCP), all of that would break

Q: What is the default gateway and why does every device need one?

- The gateway is the device's "door out" of its own subnet
- Without it, a device can only talk to others on the exact same subnet
- Inside devices point to 192.168.1.1 (the ASA); outside servers point to 5.5.5.1 (the ISP router)

3. NAT and PAT

Q: What is NAT?

- Network Address Translation
- Converts a private (internal) IP address into a public (external) IP address, and back
- Needed because private IPs (like 192.168.x.x) are not routable on the real internet

Q: What is the difference between NAT and PAT?

- NAT (static): one private IP maps to exactly one public IP — always the same, one-to-one
- PAT (dynamic/PAT overload): many private IPs share one public IP, distinguished by different port numbers — many-to-one

Q: Why does the Internal Web Server use static NAT (192.168.1.2 → 6.6.6.3) instead of PAT?

- Because outside users need to reach it at a fixed, predictable address
- PAT only works for traffic initiated from inside — it cannot be used to let outside users initiate a new connection in
- A server that needs to be reached from outside must have a static, one-to-one mapping

Q: Why do PC0/PC1/PC2 use PAT instead of static NAT?

- They are regular client devices, not servers
- They only need to initiate outbound connections (browsing, pinging), never receive unsolicited inbound ones
- PAT lets many clients share a single public IP (6.6.6.2) efficiently — this is exactly how real home/office routers work

Q: In real life, is PAT common?

- Yes — this is exactly what your home Wi-Fi router does
- Every device in your house shares your one public IP from your ISP, using PAT

Q: What command shows active NAT/PAT translations?

- `show xlate`

Q: What command shows the NAT rules that are configured?

- `show nat`

4. Access Control Lists (ACLs)

Q: What is an ACL and why is it needed?

- Access Control List — a set of rules that permit or deny specific traffic
- Needed to enforce fine-grained security policy beyond the ASA's default security-level behavior

Q: What is the "implicit deny" rule in ACLs?

- Every ACL ends with an invisible final rule: deny everything else
- So if traffic doesn't match any permit line, it is automatically dropped

Q: Why is there a separate ACL for outside (OUTSIDE-IN) and inside (INSIDE-IN)?

- OUTSIDE-IN: controls what outside users are allowed to send INTO the network (applied inbound on the outside interface)
- INSIDE-IN: controls what inside users are allowed to send OUT (applied inbound on the inside interface)
- Each interface needs its own ACL because traffic direction and trust level are different

Q: What does the OUTSIDE-IN ACL do in this project?

- Permits outside users to reach the Internal Web Server on port 80 (6.6.6.3)
- Permits ICMP (so ping replies can pass back through)
- Permits TCP/UDP replies back to the firewall's PAT address (6.6.6.2), so inside users' web/DNS sessions get their replies

Q: What does the INSIDE-IN ACL do in this project?

- Blocks all traffic (TCP, UDP, ICMP) specifically from PC0 (192.168.1.5)
- Permits all traffic from every other inside host

Q: Why did you block PC0 using TCP/UDP/ICMP deny rules instead of one "deny ip" rule?

- The simulator (Packet Tracer) rejected the "ip" keyword in that specific ACL syntax
- So the same result was achieved using three separate protocol-specific rules instead (functionally identical)

Q: How do you verify an ACL is actually being used/working?

- `show access-list` — shows each rule with a "hitcnt" (hit count)
- A rising hit count on a specific line proves real traffic is matching that rule

Q: What is ACL rule order and why does it matter?

- ACL rules are read top-to-bottom; the first match wins
- A broad "deny" placed above a specific "permit" would block that traffic before it ever reaches the permit line
- This is why deny rules for PC0 are placed before the general permit rules

5. DNS Configuration

Q: Why is a DNS server needed in this project?

- To let users type a human-friendly name (webserver.local) instead of remembering an IP address (5.5.5.3)

Q: What record type was configured, and what does it do?

- An "A Record" (Address Record)
- Maps a hostname directly to an IPv4 address: webserver.local → 5.5.5.3

Q: How do inside PCs know which DNS server to use?

- Configured via the ASA's DHCP option: `dhcpd dns 5.5.5.2 interface inside`
- So every PC that gets an IP via DHCP also automatically receives 5.5.5.2 as its DNS server

Q: How do you test DNS resolution without opening a browser?

- `nslookup webserver.local` from a PC's command prompt — shows exactly what IP the name resolves to

6. Firewall / Traffic Inspection Concepts

Q: What is "stateful inspection" and why does it matter here?

- A firewall feature where, once a connection is initiated from a trusted (inside) interface, the return traffic is automatically allowed back — without needing a separate explicit rule
- This is standard, expected ASA behavior

Q: Why was `inspect icmp` needed if ASA is supposed to be stateful by default?

- ICMP has no real "connection" the way TCP does, so the ASA needs to be told explicitly to track ICMP request/reply pairs
- Without `inspect icmp`, ping requests go out fine, but the replies get dropped on the way back

Q: Why did HTTP and DNS still fail even after NAT/ACLs looked correct?

- The simulated firewall did not properly track return traffic for TCP/UDP sessions that went through PAT
- The fix was to add an explicit ACL rule permitting TCP/UDP replies back to the PAT address (6.6.6.2) on the outside interface
- This is a simulator-specific limitation, not a real ASA design flaw

Q: What is the purpose of a default route on the ASA?

- Tells the ASA where to send any traffic whose destination isn't directly connected
- `route outside 0.0.0.0 0.0.0.0 6.6.6.1` — "anything unknown, send to the ISP router"

7. Testing and Verification

Q: How did you verify NAT/PAT was working?

- `show xlate` — showed active static and dynamic translation entries

Q: How did you verify the ACLs were actually enforcing the policy?

- `show access-list` — hit counters increased on the correct permit/deny lines during testing

Q: How did you test connectivity requirements?

- `ping` for ICMP reachability
- Web browser (http://...) for HTTP access
- `nslookup` for DNS resolution
- `telnet <ip> 80` to test raw TCP connectivity without relying on the browser

Q: Why test from more than one inside PC (PC1 and PC2)?

- To prove the configuration works generally for all inside hosts, not just by coincidence for one specific device

8. Big-Picture / "Why" Questions

Q: In one sentence, what does this whole project demonstrate?

- How a firewall uses NAT/PAT to let private devices reach the internet, and ACLs to control exactly which traffic is allowed in each direction.

Q: Why not just allow all traffic everywhere — what's the risk?

- Without restrictions, any outside attacker could directly reach every internal device
- ACLs and NAT/PAT together enforce the principle of "least privilege" — only exposing exactly what's needed (the web server), nothing else

Q: Why can outside users reach the Internal Web Server but not, say, PC1 directly?

- Only 6.6.6.3 (mapped to the web server) has a static NAT entry and a matching ACL permit
- PC1/PC2 only have dynamic PAT (outbound-only) — there is no static mapping or ACL rule allowing anything to reach them from outside, so they stay invisible/unreachable from the outside network

Q: What real-world role does this setup resemble?

- A small company's internet gateway: employees can browse the internet freely, the company website is publicly reachable, but nothing else on the internal network is exposed

Q: What would happen if you removed the static NAT for the web server?

- Outside users could no longer reach 6.6.6.3 at all, since there would be no translation from that public address to 192.168.1.2

Q: What would happen if you removed the OUTSIDE-IN ACL entirely (access-group not applied)?

- All traffic from outside would be blocked, since ASA's default behavior for low→high security level traffic is deny unless explicitly permitted — even the static NAT wouldn't help without a matching ACL permit

Quick Recap Table (memorize this)

Requirement	Feature Used	Key Command
Hide private IPs	Dynamic PAT	<code>nat (inside,outside) dynamic interface</code>
Publish internal server	Static NAT	<code>nat (inside,outside) static 6.6.6.3</code>
Allow ping both ways	ICMP inspection	<code>inspect icmp</code>
Allow outside → web server	ACL (outside)	<code>access-list OUTSIDE-IN permit tcp any host 6.6.6.3 eq 80</code>
Allow inside → outside replies	ACL (outside)	<code>access-list OUTSIDE-IN permit tcp/udp any host 6.6.6.2</code>
Block one PC's internet	ACL (inside)	<code>access-list INSIDE-IN deny ... host 192.168.1.5</code>
Name-based access	DNS A Record	<code>webserver.local → 5.5.5.3</code>